

Data Structure Visualization

Interactive animations powered by p5.js

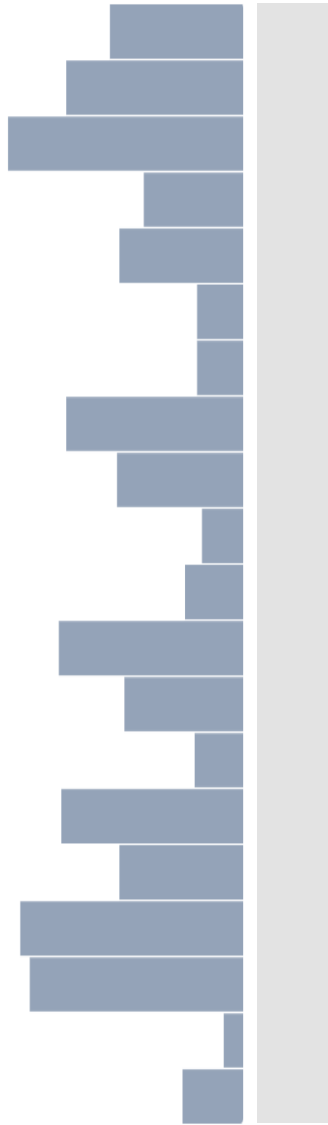


Team members:

- Benjamin Kuo
- Aman

Outline

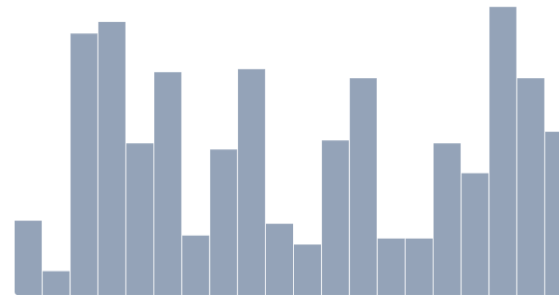
- ◇ Problem Statement
- ◇ Project Overview
- ◇ Module Breakdown
- ◇ Demonstration
- ◇ Design Highlights
- ◇ Future Extensions



Problem Statement

Why Understanding Data Structures Is Difficult ?

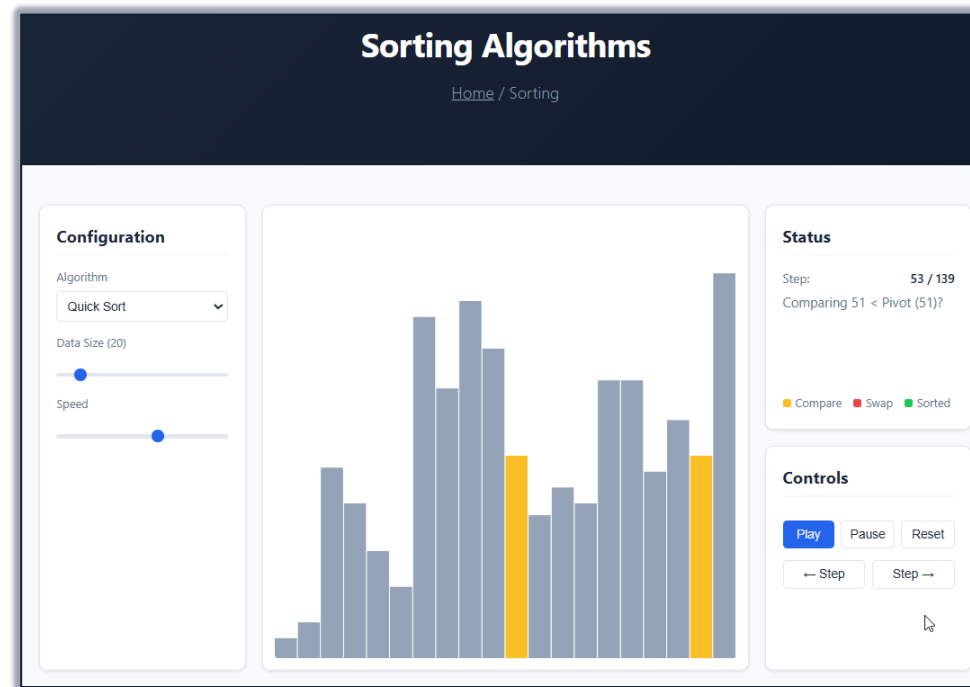
- **State changes are invisible.**
Students cannot see how operations (push, pop, insert, compare, swap) update internal structures.
- **Static diagrams can't show dynamics.**
Traditional materials fail to represent step-by-step behaviors such as pointer movement or traversal paths.



Problem Statement

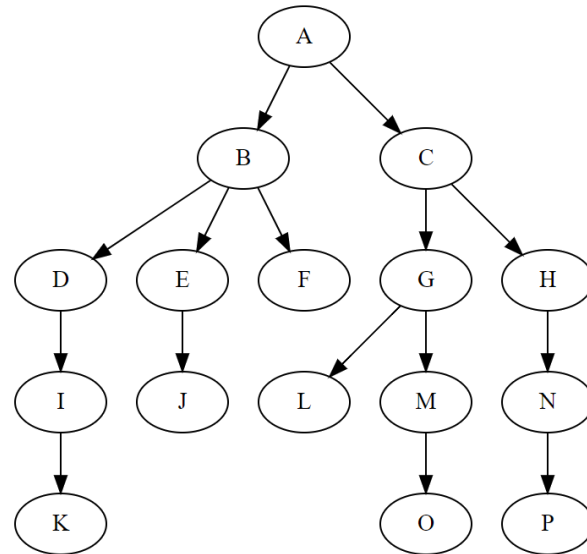
Solution

- **Make it move.**
Visual steps reveal the logic instantly.



Problem Statement

Before



Pre-order:

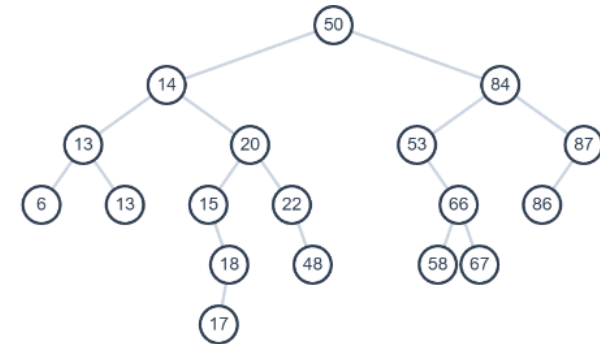
ABDIKEJFCGLMOHNP

Post-order:

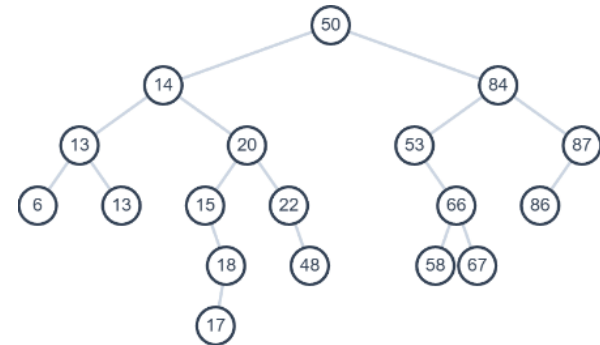
KIDJEFBLOMGPNHCA

After

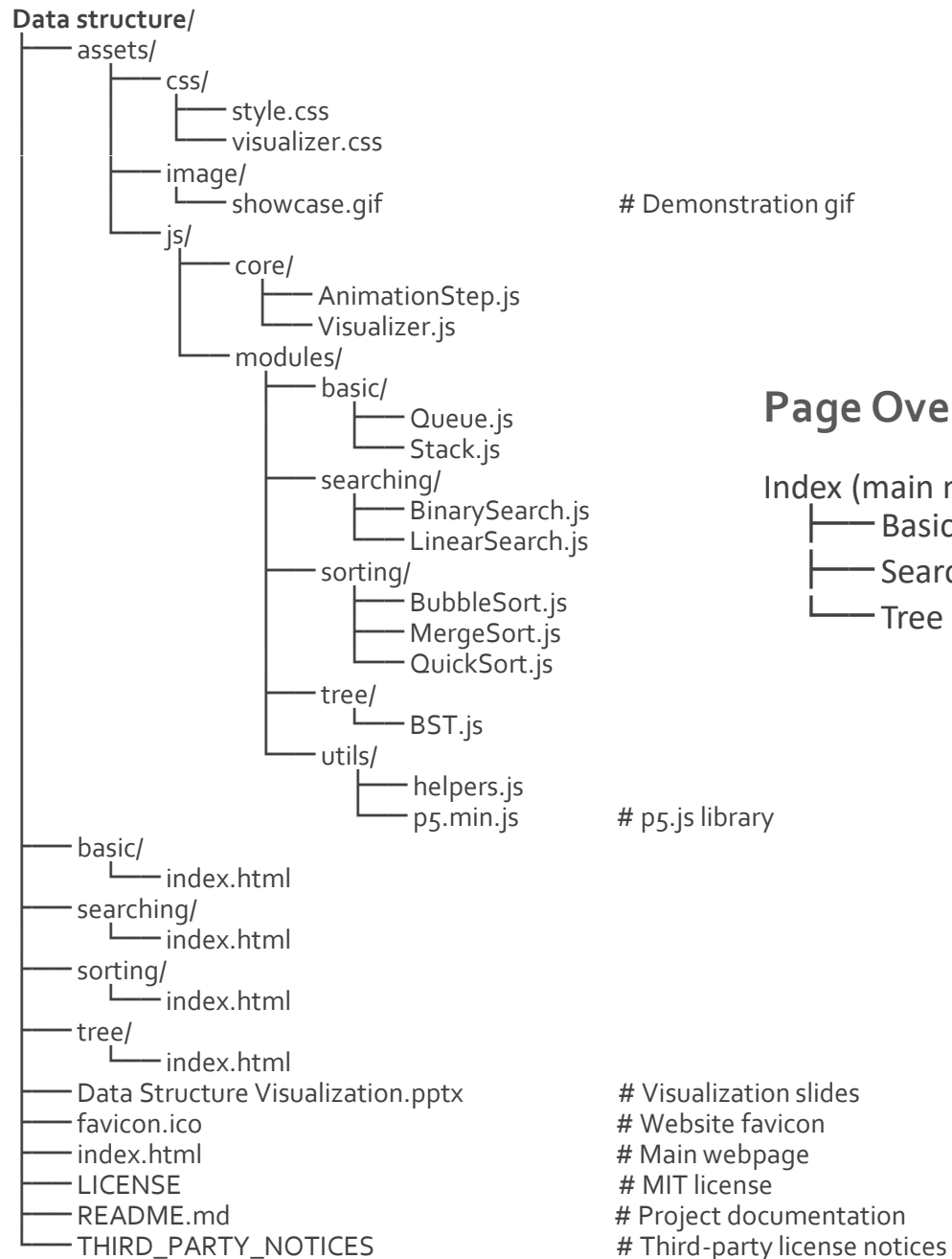
Pre-order:



Post-order:



Project Overview



Project Overview

Data Structure Visualization

Interactive animations powered by p5.js



Basic Structures

Stack · Queue · Array · Linked List



Searching

Linear · Binary · Interpolation



Sorting

Bubble · Quick · Merge · Heap

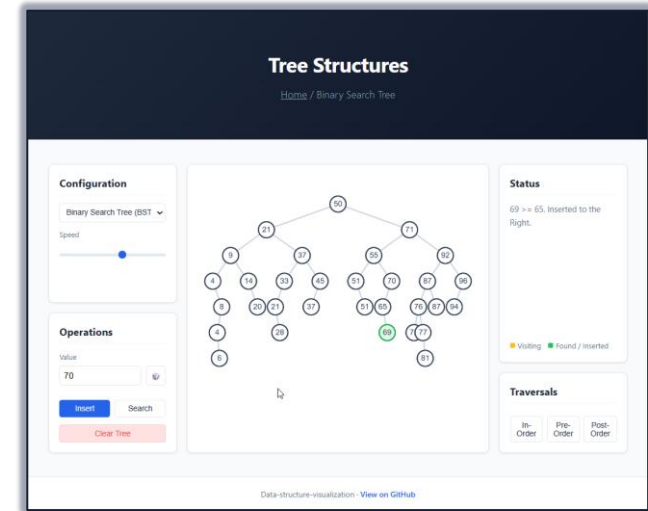
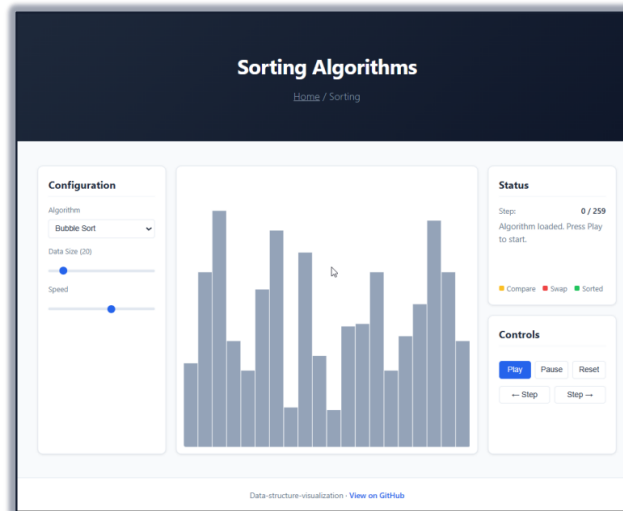
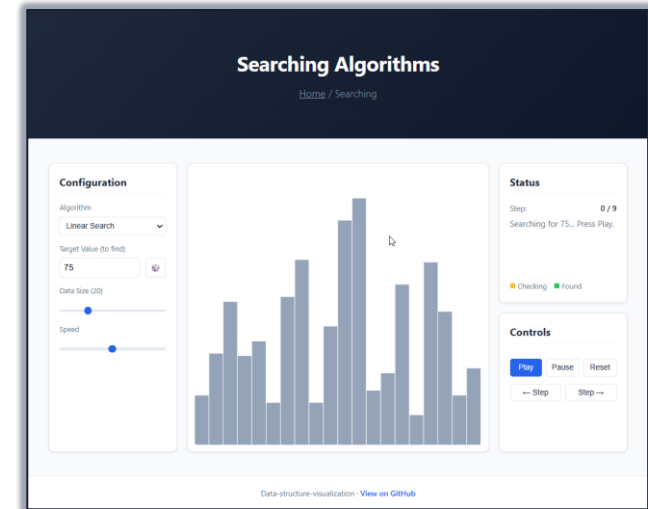
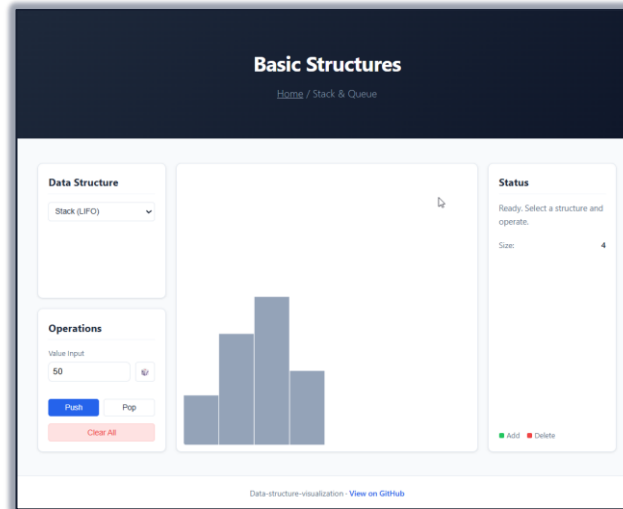


Tree Structures

BST · Heap · Traversals

Data-structure-visualization · [View on GitHub](#)

Project Overview

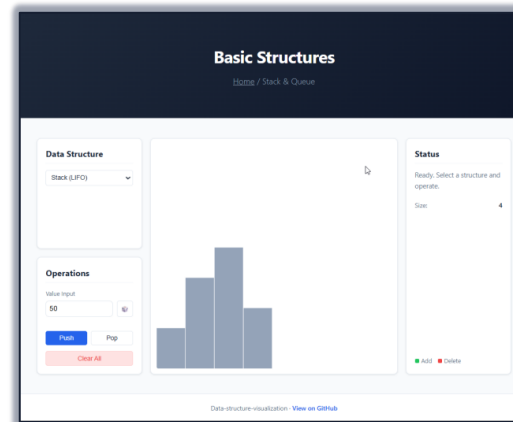


Module Breakdown

Basic Structures

Basic Structures

- **Stack** (LIFO)
Elements are added and removed from the top.
- **Queue** (FIFO)
Elements enter from the back and exit from the front.

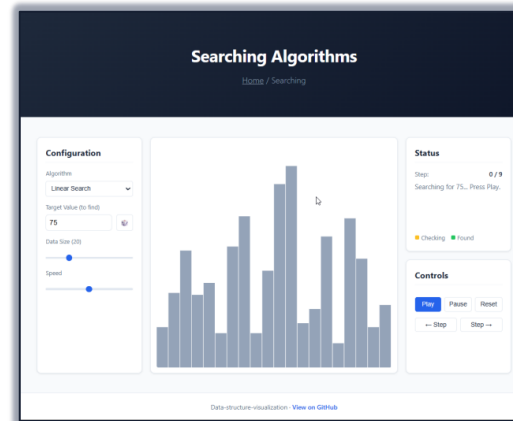


Module Breakdown

Searching

Searching

- **Linear Search**
Checks each element one by one.
- **Binary Search**
Repeatedly halves the search range on a sorted array.

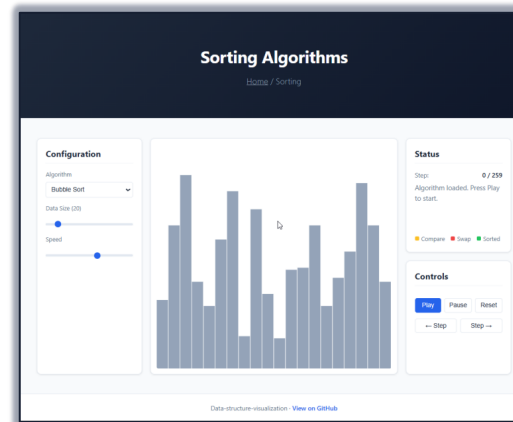


Module Breakdown

Sorting

Sorting

- **Bubble Sort**
Repeatedly compares and swaps adjacent elements.
- **Quick Sort**
Uses a pivot to partition the array.
- **Merge Sort**
Divides the array, then merges sorted halves.

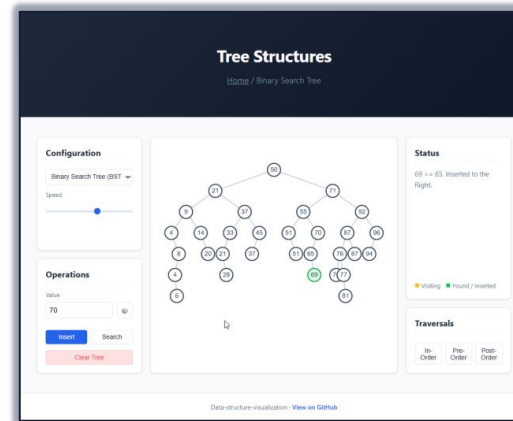


Module Breakdown

Tree Structures

Tree Structures

- **Binary Search Tree (BST)**
Values smaller go left, larger go right.
- **Traversals**
Supports Pre-order, In-order, and Post-order.



Demonstration

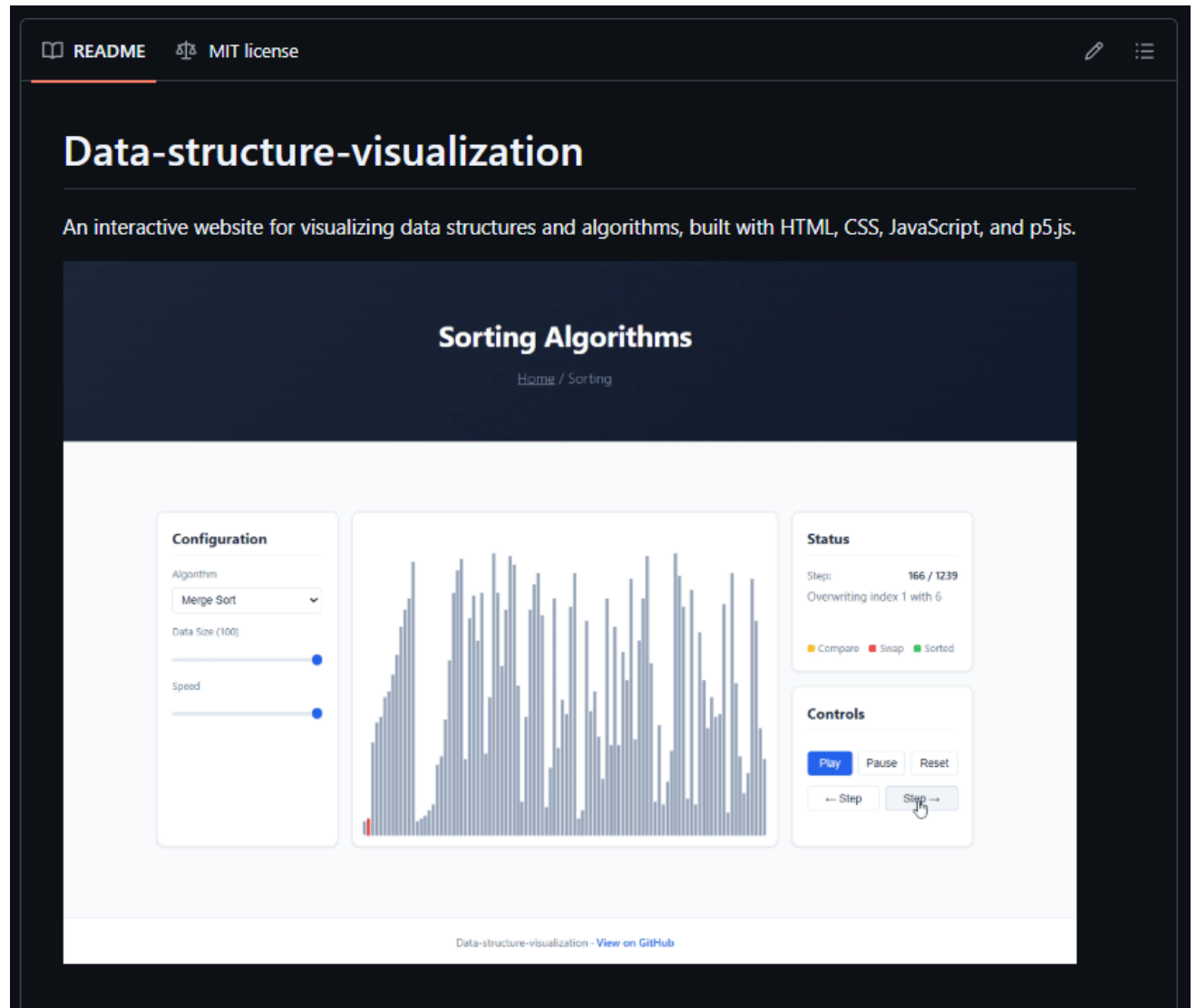
Visit the website:

<https://benjikuo.github.io/Data-structure-visualization/>

Deployments 36

✓ github-pages last week

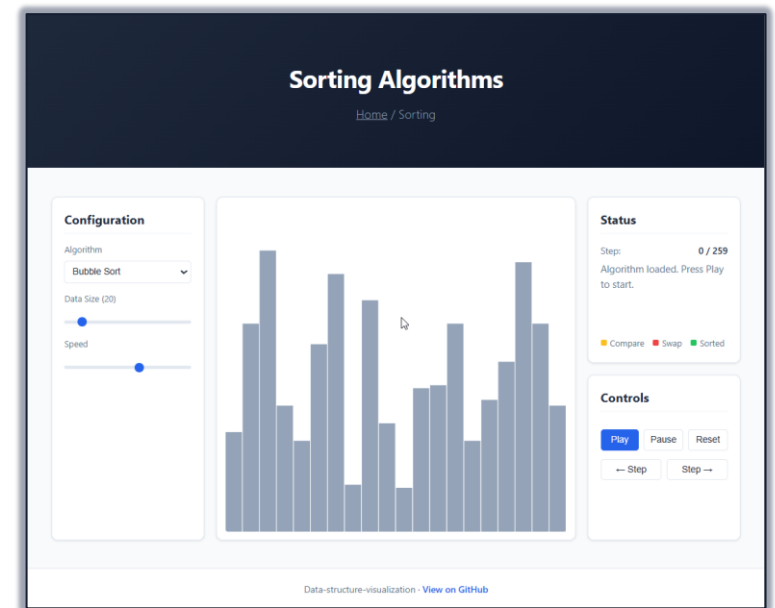
[+ 35 deployments](#)



Design Highlights

Key Design Features

- Consistent Layout
- Color-Coded States
- Interactive Controls
- Real-Time Visualization
- Modular Architecture



Future Extensions

Future Improvements

- **More Algorithms**
Add Heap Sort, Shell Sort, and other modules.
- **Add Tree Structures**
Support AVL and Red–Black Trees with rotation animations.
- **Hash Table Visualization**
Show hashing, collisions, and probing strategies.
- **Graph Algorithms**
Visualize BFS, DFS, Dijkstra, and MST on a graph canvas.



Thank You

